Δ.72 Coherence Engine, Open Data Proof of Concept

Company: Amazon (AMZN)

KPI proxies: Fulfillment expense and Shipping costs, each expressed as a percent of net sales, extracted from public SEC filings.

What this POC demonstrates

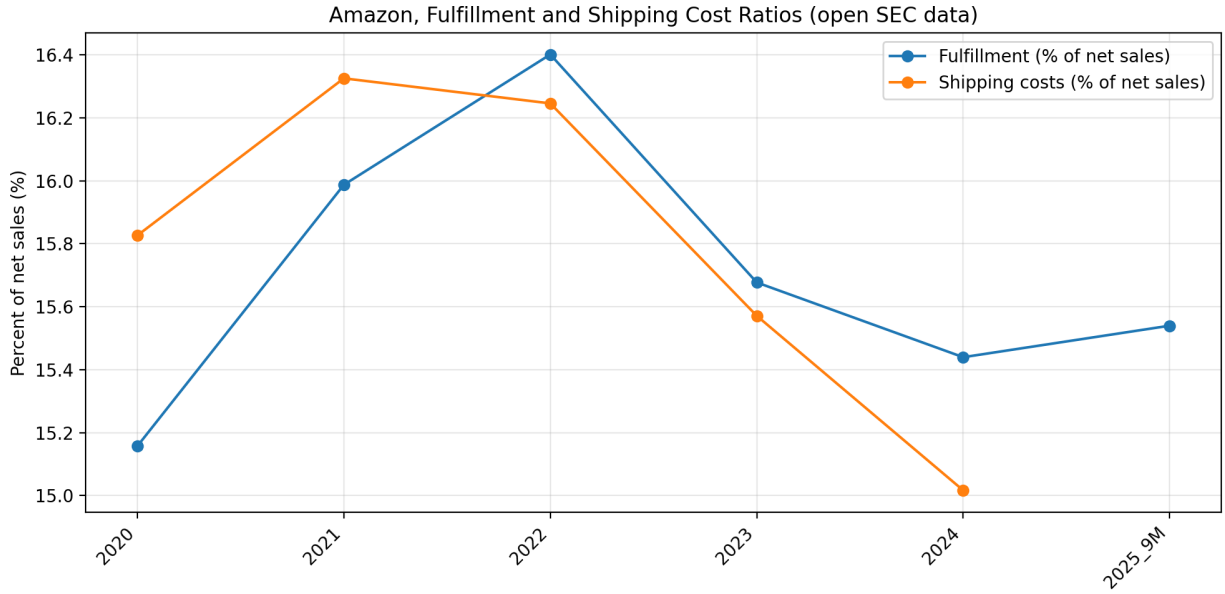
In public filings, companies rarely disclose refunds and chargebacks as a clean standalone KPI. For Amazon, the closest, consistently disclosed open proxies for operational leakage are Fulfillment expense and Shipping costs. We treat these as a measurable surface for rework, returns handling, expedites, and logistics complexity, then apply a transparent Δ.72 style coherence band to estimate potential savings during spike periods.

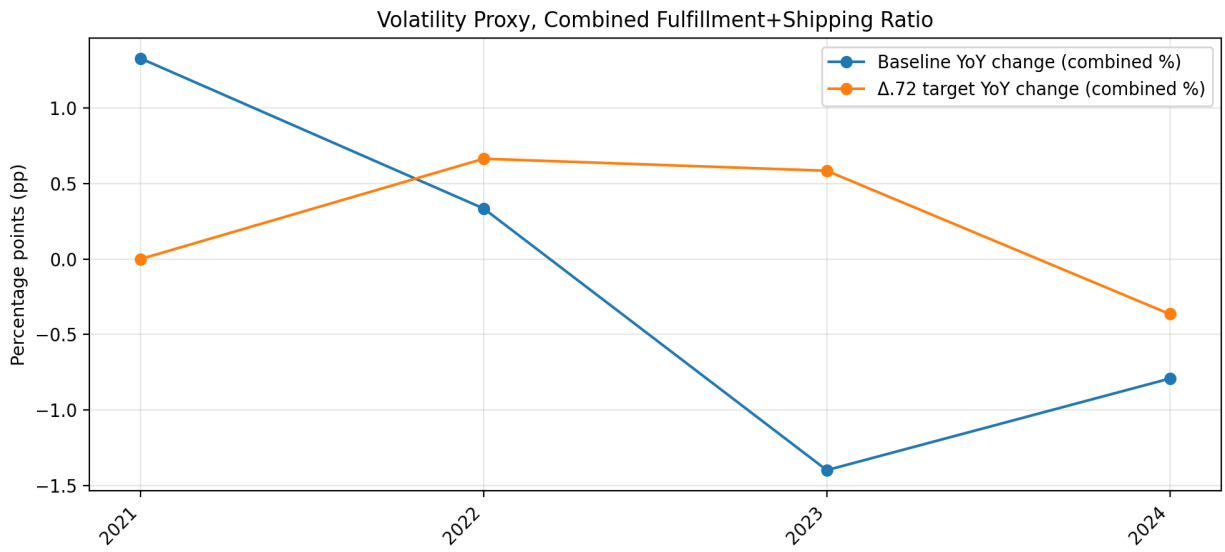
Open data used (USD, millions)

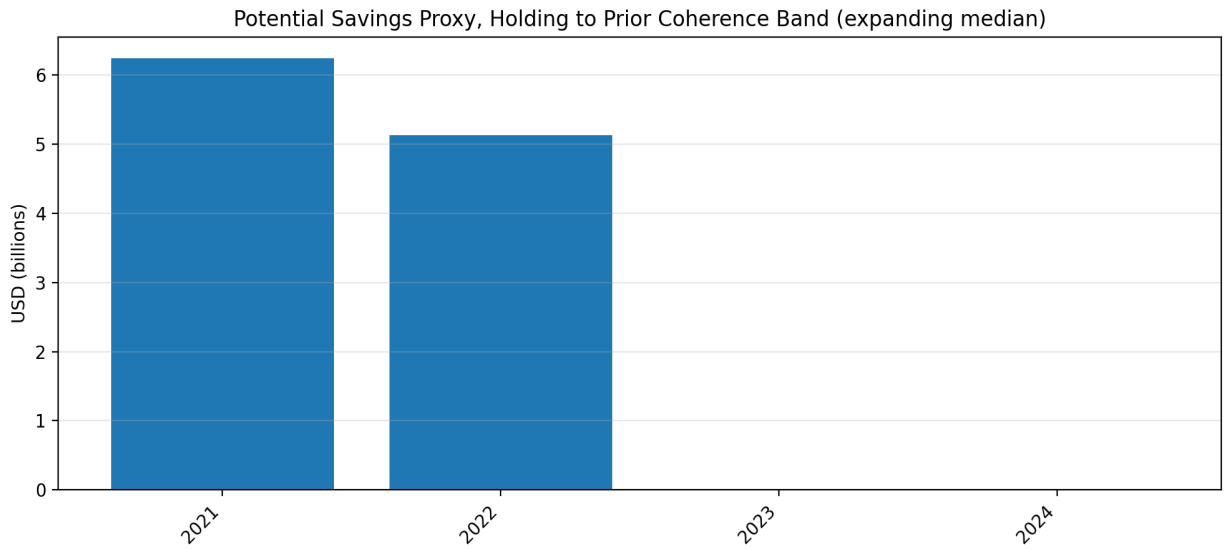
Period	Net sales (M)	Fulfillment (M)	Shipping (M)	Fulfillment %	Shipping %
2020	386064	58517	61100.0	15.16	15.83
2021	469822	75111	76700.0	15.99	16.33
2022	513983	84299	83500.0	16.4	16.25
2023	574785	90109	89500.0	15.68	15.57
2024	637959	98498	95800.0	15.44	15.02
2025_9M	503538	78248		15.54	

Note: 2025_9M is the nine months ended September 30, 2025, included to extend the timeline with the latest in-range open data.

Charts




Volatility Proxy, Combined Fulfillment+Shipping Ratio
Baseline YoY change (combined %)
Δ.72 target YoY change (combined %)
Percentage points (pp)
1.0
0.5
0.0
−0.5
−1.0
−1.5
2021
2022
2023
2024


Potential Savings Proxy, Holding to Prior Coherence Band (expanding median)
USD (billions)
6
5
4
3
2
1
0
2021
2022
2023
2024

Savings methodology (transparent proxy)

For each year t, define a coherence target band as the expanding median of prior years for each ratio. Potential savings is computed when the observed ratio exceeds the prior median, multiplied by net sales. This yields an interpretable, conservative estimate of the dollar impact of holding operational leakage to a historically coherent level during spike years.

Backtest table (ratio targets, savings proxy)

Period	Fulfill %	Target %	Ship %	Target %	Savings (M)
2020	15.16		15.83		
2021	15.99	15.16	16.33	15.83	6242.7
2022	16.4	15.57	16.25	16.08	5133.2
2023	15.68	15.99	15.57	16.25	0.0
2024	15.44	15.83	15.02	16.04	0.0
2025_9M	15.54	15.68		15.83	

∆.72 summary block (Amazon)

Metric	Value
baseline_mean_combined_pct_2020_2024	31.53%
delta72_target_mean_combined_pct_2020_2024	31.54%
efficiency_uplift_frac_spike_window	3.56%
variance_reduction_frac_yoy_changes	83.59%
recovery_time_years_from_peak	1
thd_proxy_baseline	0.0844
thd_proxy_delta72	0.0743
total_potential_savings	$11.38B

Important: This is a proof-of-concept using public, aggregate disclosures. A production deployment would ingest internal operational data (returns rate, chargebacks, refunds, expedite spend, rework tickets) to calculate a direct coherence score and savings attribution with audit trails.